which the stochastic terms are eliminated and replaced by a kinetic energy control law based only on relative velocities of interacting pairs. We refer to this as the pairwise Nosé-Hoover (PNH) thermostat; it is designed to conserve the momentum and is also Galilean-invariant. This method is particularly useful in nonequilibrium molecular dynamics (NEMD) applications where the total momentum needs to be preserved under thermostatting [342].

The equations of motion of the PNH thermostat (for particle i) is given by

$$\begin{aligned} \mathrm{d}\boldsymbol{q}_i &= m_i^{-1}\boldsymbol{p}_i \mathrm{d}t, \\ \mathrm{d}\boldsymbol{p}_i &= \boldsymbol{F}_i^{\mathrm{C}}(\boldsymbol{q})\mathrm{d}t + \xi\tilde{\boldsymbol{F}}_i^{\mathrm{D}}\mathrm{d}t, \\ \mathrm{d}\xi &= G(\boldsymbol{q},\boldsymbol{p})\mathrm{d}t, \end{aligned} \tag{8.108}$$

where ξ is a dynamical variable, the dissipative/driving coupling term is defined as

$$\tilde{\boldsymbol{F}}_i^{\mathrm{D}} = \sum_{j=1}^{\mathrm{N}} \tilde{\boldsymbol{F}}_{ij}^{\mathrm{D}}, \qquad \tilde{\boldsymbol{F}}_{ij}^{\mathrm{D}} = -\omega^{\mathrm{D}}(r_{ij})(\hat{\boldsymbol{q}}_{ij}\cdot\boldsymbol{v}_{ij})\hat{\boldsymbol{q}}_{ij},$$

(i.e. $\gamma = 1$ in the dissipative term for DPD), and $G(\boldsymbol{q},\boldsymbol{p})$ is the instantaneous accumulated deviation of the kinetic temperature away from the target temperature [8]

$$G(\boldsymbol{q},\boldsymbol{p}) = \mu^{-1}\sum_i\sum_{j>i}\omega^{\mathrm{D}}(r_{ij})\left[\left(\boldsymbol{v}_{ij}\cdot\hat{\boldsymbol{q}}_{ij}\right)^2 - \mathrm{k_B T}/m_{ij}\right], \tag{8.109}$$

where μ is a coupling parameter which is referred to as the "thermal mass". The invariant distribution is defined in the $(\boldsymbol{q},\boldsymbol{p},\xi)$ extended space, with density

$$\hat{\rho}_\beta(\boldsymbol{q},\boldsymbol{p},\xi) = \hat{Z}^{-1}\exp\left(-\beta H(\boldsymbol{q},\boldsymbol{p})\right)\exp\left(-\beta\mu\xi^2/2\right), \tag{8.110}$$

where $\hat{Z}$ is a suitable normalizing coefficient so that the density integrates to one over the entire accessible space (subject, as usual, to the conservation of momentum constraint).

A nonsymmetric integration algorithm was applied in the original paper [8] to solve the system.

8.9.4 Pairwise Nosé-Hoover-Langevin Thermostat

To improve ergodicity, it is intuitive to employ the same stochastic device used in the Nosé-Hoover-Langevin method, namely linear (OU-type) stochastic terms introduced in the equation for the auxiliary variable ξ. If the OU process is properly tuned the extended canonical distribution with density $\hat{\rho}_\beta$ will still be preserved.